

Supplementary Material

Walk-unbiased doubly robust graph propagation for recommendation from exposure-biased logs

Additional results referred to in the main text. Theorem, table and section numbers without the prefix S refer to the main text.

Table S1: Stress tests outside Assumption 1 and with misspecified propensities (log-normal noise with $\sigma = 0.5$). Relative $|\text{bias}|$ on unexposed candidates.

Scenario ($K = 3$, rel. $ \text{bias} $ on unexposed candidates)	IPS	IPS + WC	DR	DRUP
independent exposures (Assumption 1)	0.029	0.029	1.137	0.019
fixed-size slates per user (Proposition 4)	0.219	0.219	1.119	0.024
misspecified propensities, exact imputation	0.568	0.568	0.000	0.000
misspecified propensities, noisy imputation	0.568	0.568	1.916	0.158
independent exposures (Assumption 1), $K = 5$	4.667	0.059	5.053	0.040
fixed-size slates per user (Proposition 4), $K = 5$	2.392	0.416	4.633	0.032
misspecified propensities, noisy imputation, $K = 5$	19.266	1.162	16.027	0.204

Table S2: Nuisance protocol in the 40-user, 60-item model of Table 3 with true and with estimated (logistic exposure model, misspecified here) propensities, nuisances from an independent log, cross-fitted, or fitted on the same log without cross-fitting, and degrees from $\hat{Y}$ or from edge estimates. Relative $|\text{bias}|$ of the three-hop term and of the full score with data-dependent constants (3,000 replications).

Propensities	Nuisances	degrees	IPS edges			DR edges		
			uncorr.	DRUP	$s, \text{ data } c$	uncorr.	DRUP	$s, \text{ data } c$
true	independent log	$\hat{Y}$	2.15	0.10	0.05	1.90	0.09	0.35
	independent log	W	5.04	0.25	0.10	3.59	0.18	0.35
	cross-fitted	$\hat{Y}$	0.39	0.64	0.15	0.61	0.80	0.62
	cross-fitted	W	3.12	0.81	0.11	4.11	0.73	0.49
	same log	$\hat{Y}$	0.42	0.70	0.18	0.59	0.70	0.33
	same log	W	0.94	0.97	0.49	0.80	0.86	0.37
estimated	independent log	$\hat{Y}$	29.57	3.70	0.15	25.57	2.11	0.53
	independent log	W	92.66	7.44	0.15	47.74	3.00	0.50
	cross-fitted	$\hat{Y}$	1.29	0.48	0.11	3.41	1.23	0.92
	cross-fitted	W	83.73	0.49	0.18	27.63	0.73	0.59
	same log	$\hat{Y}$	0.60	0.82	0.21	0.64	0.72	0.33
	same log	W	0.93	0.97	0.53	0.83	0.88	0.38

Table S3: Tightness of Theorem 6 in a 12-user, 15-item model (20,000 replications, all propensities $\geq \tau$), over all (u, i) .

	$\tau = 0.3$	$\tau = 0.1$
variance bound (b) / Monte-Carlo variance, median	1582	8422
variance bound (b) / Monte-Carlo variance, minimum	471	2126
McDiarmid half-width (d) / empirical 95% quantile, median	131	1831
McDiarmid half-width (d) / empirical 95% quantile, minimum	73	685

Table S4: Exposure-capped re-ranking for all caps. Cells: nDCG@ K (Gini@ K).

Coat	none	$c = 20$	$c = 10$	$c = 5$	$c = 3$	$c = 2$	$c = 1.5$
logged graph	0.502 (0.34)	0.494 (0.34)	0.501 (0.34)	0.501 (0.34)	0.493 (0.33)	0.494 (0.32)	0.487 (0.28)
IPS adjacency	0.507 (0.35)	0.497 (0.35)	0.506 (0.35)	0.503 (0.34)	0.504 (0.36)	0.493 (0.33)	0.486 (0.29)
DR adjacency	0.489 (0.60)	0.489 (0.60)	0.489 (0.60)	0.489 (0.60)	0.485 (0.58)	0.475 (0.48)	0.444 (0.37)
DRUP	0.489 (0.60)	0.489 (0.60)	0.489 (0.60)	0.489 (0.60)	0.485 (0.58)	0.476 (0.48)	0.443 (0.37)
KuaiRec	none	$c = 20$	$c = 10$	$c = 5$	$c = 3$	$c = 2$	$c = 1.5$
logged graph	0.603 (0.99)	0.300 (0.91)	0.228 (0.84)	0.173 (0.72)	0.139 (0.60)	0.115 (0.44)	0.102 (0.31)
IPS adjacency	0.612 (0.99)	0.307 (0.94)	0.234 (0.89)	0.178 (0.79)	0.146 (0.66)	0.121 (0.49)	0.107 (0.34)
DR adjacency	0.639 (0.99)	0.335 (0.95)	0.257 (0.90)	0.199 (0.80)	0.165 (0.67)	0.140 (0.50)	0.125 (0.35)
DRUP	0.638 (0.99)	0.335 (0.95)	0.256 (0.90)	0.199 (0.80)	0.164 (0.67)	0.140 (0.50)	0.125 (0.35)
Yahoo! R3	none	$c = 20$	$c = 10$	$c = 5$	$c = 3$	$c = 2$	$c = 1.5$
logged graph	0.658 (0.28)	0.659 (0.28)	0.660 (0.28)	0.658 (0.28)	0.656 (0.28)	0.658 (0.27)	0.652 (0.23)
IPS adjacency	0.656 (0.27)	0.657 (0.27)	0.657 (0.27)	0.656 (0.27)	0.656 (0.27)	0.656 (0.27)	0.651 (0.22)
DR adjacency	0.656 (0.27)	0.657 (0.27)	0.657 (0.27)	0.656 (0.27)	0.656 (0.27)	0.656 (0.27)	0.651 (0.22)
DRUP	0.656 (0.27)	0.657 (0.27)	0.657 (0.27)	0.656 (0.27)	0.656 (0.27)	0.656 (0.27)	0.651 (0.22)

Table S5: Best nDCG over each operator’s hyper-parameter grid under a cap on the Gini coefficient of top- K exposure.

Data	Operator	Gini ≤ 0.3	≤ 0.4	≤ 0.5	≤ 0.6	no cap
Coat	logged graph	–	0.492	0.492	0.492	0.492
	IPS adjacency	–	0.495	0.495	0.495	0.495
	DR adjacency	–	0.417	0.462	0.486	0.498
	DRUP	–	0.417	0.441	0.484	0.497
Yahoo! R3	logged graph	0.659	0.659	0.659	0.659	0.659
	IPS adjacency	0.657	0.657	0.657	0.657	0.657
	DR adjacency	0.657	0.657	0.657	0.657	0.657
	DRUP	0.657	0.657	0.657	0.657	0.657
KuaiRec	logged graph	–	–	–	–	0.603
	IPS adjacency	–	–	–	–	0.612
	DR adjacency	–	–	–	–	0.633
	DRUP	–	–	–	–	0.633

Table S6: Certificates under re-fitted nuisances: the nuisances are re-fitted on the poisoned log (no cross-fitting) and the realised top- K is checked against the frozen-nuisance certificates of the clean system.

Data	Operator	fake users F	hit rate (frozen / re-fitted)	certified pairs	violated
Coat	IPS	1, 2, 5, 10	0.001/0.001, 0.018/0.001, 0.268/0.001, 0.536/0.001	217	0
Coat	DR	1, 2, 5, 10	0.000/0.000, 0.000/0.000, 0.000/0.000, 0.004/0.000	5193	0
Coat	DRUP	1, 2, 5, 10	0.000/0.000, 0.000/0.000, 0.000/0.000, 0.004/0.000	5196	0
Yahoo! R3	IPS	1, 5, 20, 100	0.065/0.065, 0.065/0.065, 0.066/0.064, 0.174/0.064	0	0
Yahoo! R3	DR	1, 5, 20, 100	0.065/0.065, 0.065/0.065, 0.066/0.064, 0.174/0.064	0	0
Yahoo! R3	DRUP	1, 5, 20, 100	0.065/0.065, 0.065/0.065, 0.066/0.064, 0.174/0.064	0	0
KuaiRec	IPS	5, 50, 200	0.001/0.001, 0.001/0.001, 0.133/0.001	1616	0
KuaiRec	DR	5, 50, 200	0.000/0.000, 0.000/0.000, 0.500/0.000	2819	0
KuaiRec	DRUP	5, 50, 200	0.000/0.000, 0.000/0.000, 0.500/0.000	2821	0

Table S7: Jointly private DRUP for all fixed denoising ranks ($\delta = 10^{-5}$).

Data	rank	$\epsilon = 0.5$	1	2	4	8	16	∞
Coat	8	0.397	0.403	0.430	0.441	0.455	0.459	0.452
	32	0.391	0.402	0.416	0.434	0.454	0.457	0.454
	128	0.370	0.390	0.412	0.440	0.436	0.452	0.453
	full	0.380	0.398	0.409	0.435	0.445	0.457	0.455
	non-private DRUP (all nuisances) 0.555; popularity 0.527; 29 public users							
Yahoo! R3	8	0.387	0.445	0.491	0.522	0.558	0.581	0.597
	32	0.397	0.437	0.480	0.518	0.551	0.574	0.608
	128	0.397	0.433	0.472	0.505	0.535	0.561	0.610
	full	0.399	0.428	0.470	0.496	0.526	0.553	0.608
	non-private DRUP (all nuisances) 0.652; popularity 0.525; 1540 public users							
KuaiRec	8	0.055	0.060	0.071	0.185	0.541	0.598	0.635
	32	0.049	0.052	0.055	0.163	0.530	0.596	0.635
	128	0.049	0.049	0.048	0.133	0.484	0.583	0.634
	full	0.050	0.054	0.061	0.089	0.163	0.316	0.635
	non-private DRUP (all nuisances) 0.637; popularity 0.577; 717 public users							